

MAT1100

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1 Group Theory

1.1 Basic Definitions and Results

Definition 1. A Group is a set with a binary operation $\star : G \times G \rightarrow G$ satisfying

1. Associativity: for all $a, b, c \in G$, $(a \star b) \star c = a \star (b \star c)$
2. Existence of an identity: $\exists e \in G$ such that for all $a \in G$, $a = a \star e = e \star a$.
3. Existence of inverses: for all $a \in G$ there exists $b \in G$ such that $ab = ba = e$.

Example 1. Some examples

1. $(\mathbb{Z}, +)$
2. $GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) \neq 0\}$
3. $SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) = 1\}$

Example 2. D_8 the group of symmetries of the unit square S .

$$D_8 = \{T \in GL_2(\mathbb{R}) : T(S) = S \wedge \|T(x) - T(y)\| = \|x - y\|, \forall x, y \in \mathbb{R}^2\}$$

it is not hard to check that this forms a group under function composition.

Definition 2. Subgroups $H \subset G$ is a subgroup and denoted $H \leq G$ if it is a group under the operation of G .

Equivalently:

1. $e \in H$
2. $xy \in H$ for all $x, y \in H$
3. $x^{-1} \in H$ for all $x \in H$.

Remark 1. There is a subgroup criterion that is sometimes useful: $H \subseteq G$ is a subgroup if and only if $H \neq \emptyset$ and for all $x, y \in H$, $xy^{-1} \in H$

Example 3. Some examples

1. $SL_n(\mathbb{K}) \leq GL_n(\mathbb{K})$
2. $(\mathbb{Z}, +) \leq (\mathbb{Q}, +)$
3. $D_8 \leq GL_2(\mathbb{R})$.

Definition 3. For $g \in G$ a group, the cyclic subgroup generated by g is

$$\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$$

Example 4. $S_3 := \{\sigma : \{1, 2, 3\} \rightarrow \{1, 2, 3\} \text{ a bijection}\}$

$$\langle (1, 2) \rangle = \{e, (1, 2)\}$$

and

$$\langle (1, 2, 3) \rangle = \{e, (1, 2, 3), (1, 3, 2)\}$$

Definition 4. The order of a group G is its cardinality and denoted $|G|$. For $g \in G$, $|g| = \min\{n \in \mathbb{N} : g^n = e\}$ called the order of g .

Proposition 1. For $g \in G$ we have $|\langle g \rangle| = |g|$.

Proof. Set $n = |g|$. By the minimality of the order $e, g, \dots, g^{n-1}$ are all distinct. Now it suffices to show that $\langle g \rangle = \{e, g, \dots, g^{n-1}\}$ the reverse containment is clear. For the forward containment, let $g^a \in \langle g \rangle$. $a \in \mathbb{Z}$, so there exists $k \in \mathbb{Z}$ and $r \in 0, \dots, n-1$ such that

$$a = kn + r$$

then

$$g^a = g^{kn+r} = g^{kn}g^r = g^r$$

hence g^a is represented by an element in the right hand set which proves the proposition. $\square$

Definition 5. A group homomorphism is a map $\varphi : G \rightarrow G'$ such that $\varphi(gh) = \varphi(g)\varphi(h)$ for all $g, h \in G$.

1. The kernel of φ : $\ker \varphi = \{g \in G | \varphi(g) = e_{G'}\}$
2. The image of φ : $\text{Im } \varphi = \{\varphi(g) \mid g \in G\}$

Proposition 2. $\ker \varphi \leq G$ and $\text{Im } \varphi \leq H$.

Example 5. $\varphi = \det : GL_n(\mathbb{K}) \rightarrow \mathbb{K}^\times$ with $\ker \varphi = SL_n(\mathbb{K})$.

Example 6. Now we compute D_8 explicitly. Label the vertices of the square 1, 2, 3, 4 starting from the bottom left and moving counter clockwise.

A symmetry of the square must be a permutation as the vertices are exactly the points of the square which have another point a distance of the diagonal away and isometries must preserve this.

Additionally isometries must preserve adjacency

Hence given an element $T \in D_8$, $T(1)$ has 4 possibilities, then given $T(1)$, $T(2)$ has 2 and then by adjacency the isometry is now determined.

Therefore $|D_8| \leq 8$.

Now it is clear that two isometries are:

1. s: reflection about the diagonal 1-3.
2. r: counter-clockwise rotation by $\frac{\pi}{2}$

and from here it is easy to show that

$$D_8 = \{e, s, r, r^2, r^3, rs, r^2s, r^3s\}$$

and from here we recover another group as a subgroup

$$\{e, s, r^2, r^2s\} = K_4$$

the Klein 4-group.